On independence tests for polyspherical data and their applications

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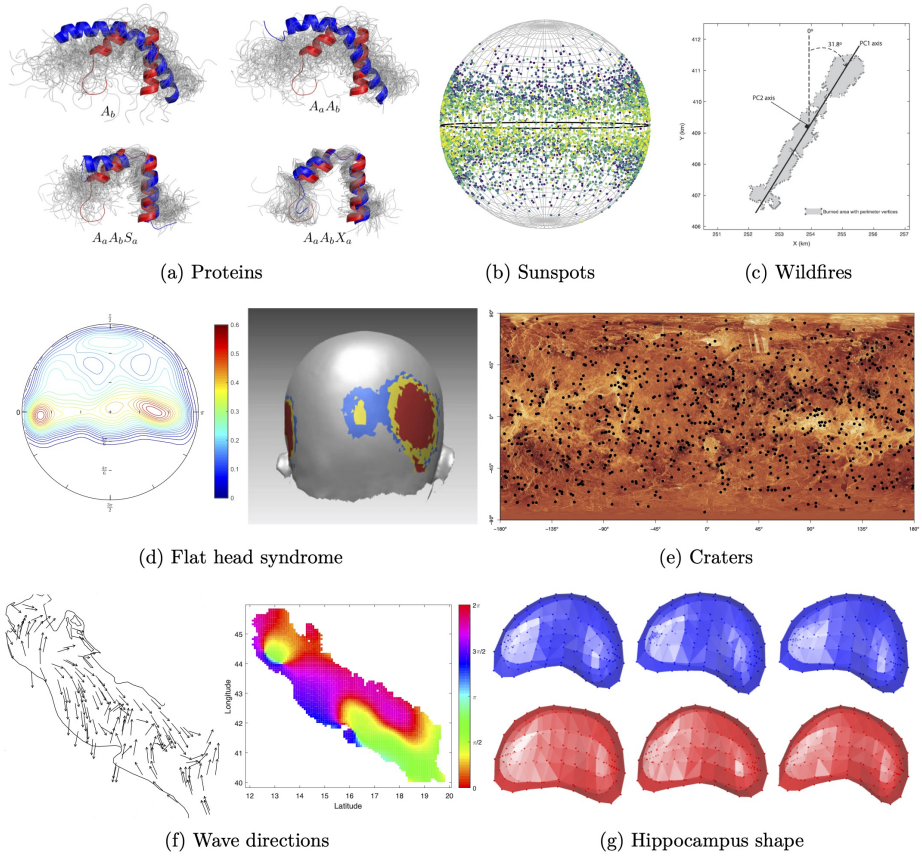
April 11, 2025

1 Motivation

Directional data refers to observations recorded as angles or directions, usually represented on the circumference, sphere, or Cartesian products thereof. We use the term *polyspherical data* to describe observations on the product space of spheres $\mathbb{S}^{d_1} \times \cdots \times \mathbb{S}^{d_r}$, where $d_1, \ldots, d_r \geq 1$ and $\mathbb{S}^d := \{\mathbf{x} \in \mathbb{R}^{d+1} : \|\mathbf{x}\| = 1\}$. The polysphere includes cases like the circle ($r = d_1 = 1$), the sphere ($r = 1$, $d_1 = 2$), and the torus ($d_1 = \cdots = d_r = 1$). Examples of polyspherical data are found in medical imaging (e.g., Vuollo et al., 2016; Schulz et al., 2016; Pizer et al., 2022), protein/RNA structures (e.g., Hamelryck et al., 2006; Boomsma et al., 2008), and astronomical studies like comet orbits and cosmic rays (e.g., Jupp et al., 2003; Baldi et al., 2009), to name but just a few. High-dimensional spherical data can emerge from standardizing multivariate data, seen in genetics (Eisen et al., 1998) and text mining (Banerjee et al., 2005). Mardia and Jupp (1999), Ley and Verdebout (2017), and Pewsey and García-Portugués (2021) detail methods and advances in directional statistics, the branch of statistics dealing with observations involving data on $\mathbb{S}^d$. Marron and Dryden (2021) offers insights into new methodologies involving directional data. Figure 1 shows several applications of directional statistics and examples of polyspherical data.

An important and recent instance of polyspherical data comprises the parameterization of three-dimensional shapes through skeletal models, or s-reps (Pizer et al., 2013; Liu et al., 2021). These models comprise a central skeleton and spokes representing the object's shape more comprehensively than boundary-only models. See Figure 2 for graphical insight on the construction of s-reps. The geometry of s-reps, especially their spokes, make them heavily connected to a polysphere, as a scaling operation to register them removes the size information and preserves only the shape. s-reps models have been successfully applied to analyzing three-dimensional shapes in medical imaging. Indeed, works such as Jung et al. (2011), Jung et al. (2012), and Pizer et al. (2013) have explored the shape of the hippocampus (see Figure 3) using methods based on s-reps. Other recent research by Schulz et al. (2015), García-Portugués and Meilán-Vila (2023), and Liu et al. (2023) has expanded on these advances for from the optic of more flexible approaches.



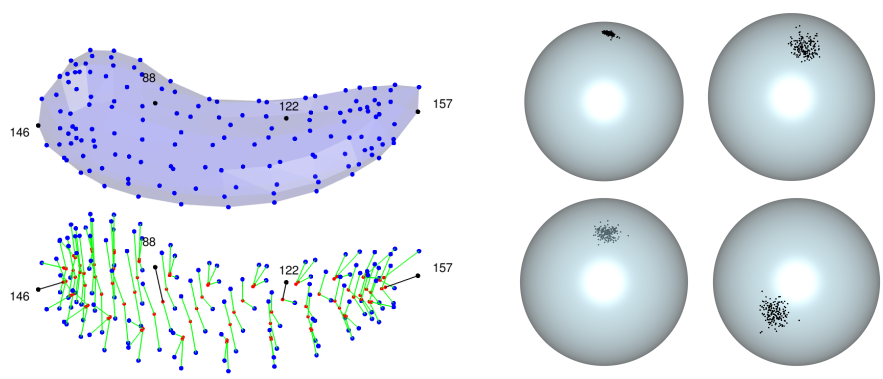
(a) Proteins (b) Sunspots (c) Wildfires

(d) Flat head syndrome (e) Craters

(f) Wave directions (g) Hippocampus shape

Figure 1: Some applications of directional statistics: (a) increasingly accurate prediction (blue ribbon) of a protein structure (red ribbon) by sampling of candidate structures (grey filaments) with a hidden Markov model using a diffusive process on $\mathbb{S}^1 \times \mathbb{S}^1$ (García-Portugués et al., 2018); (b) presumably rotationally symmetric distribution on $\mathbb{S}^2$ of the first-ever recorded positions of the sunspots of the 23rd solar cycle, with colour indicating the appearance time within the solar cycle (García-Portugués et al., 2020); (c) determination of the principal orientation (an object on $\mathbb{S}^1$ with signs identified) of a wildfire perimeter by principal component analysis (García-Portugués et al., 2014); (d) analysis of the flat head syndrome (right plot, reddish areas) using the density of the directions of the normal vectors (left plot, data on $\mathbb{S}^2$) to the cranium surface (Vuollo et al., 2016); (e) locations of craters (black dots, data on $\mathbb{S}^2$) on the colourised surface of Venus, exhibiting a very unusual and surprisingly uniform distribution (García-Portugués et al., 2021); (f) wave directions in the Adriatic Sea (left, series of observations on $\mathbb{S}^1$) and estimation of the wave directions trend (right), accounting the spatial dependence structure (Meilán-Vila et al., 2021); (g) Skeletal model reconstruction of hippocampus shapes, with determination of prototype (blue, upper row) and rare hippocampi (red, bottom row) shapes from the analysis of data on $(\mathbb{S}^2)^{168}$ (García-Portugués and Meilán-Vila, 2024).

Past research with s-reps has evidenced that some spokes are heavily correlated between them (see, e.g., García-Portugués and Meilán-Vila, 2023). Therefore, spokes at different locations are expressing information on the shape of the s-rep in a repeated way and are therefore redundant when their directions are fed into a classifier that separates ill from healthy patients. In addition, an s-rep model can have a significant number of spokes, such as $r = 168$ for the relatively simplistic hippocampus model. A test of independence between the directions at different spokes of the s-rep model that outputs a p-value in a sensible time would be a most valuable tool to quantify the amount of dependence between different spokes, thus facilitating the removal of the redundant ones to retain only those maximally informative. Furthermore, an extension of such a test to accommodate a class label or a response variable will allow scanning for the spokes that are more correlated with that target variable to include them in classifiers or predictive models. Independence tests are, in any case, fundamental tools in a statistical analysis and typically precede any modeling effort. Independence tests for polyspherical data are virtually inexistent beyond the simple $\mathbb{S}^1 \times \mathbb{S}^1$ case (Pewsey and García-Portugués, 2021) and this project aims to contribute to filling this gap of knowledge.



(a) Construction process of an s-rep. (b) Directions on $\mathbb{S}^2$ of specific spokes.

Figure 2: Construction of an s-rep and scatterplots of directions of different spokes. Figure 2a sketches the s-rep fitting process (Liu et al., 2021), where a hippocampus surface (blue, top) is parametrized as a set of base skeletal points (red) and a collection of 168 spokes (green segments) connecting with the boundary (blue points). Figure 2b shows the $\mathbb{S}^2$-directions of the $n = 177$ spokes located at boundary points 88, 122, 146, and 157 (from left to right and top to bottom). The locations of these spokes are shown in Figure 2a. Figure extracted from García-Portugués and Meilán-Vila (2024).

2 Objectives

The purpose of this project is to develop a new nonparametric test of independence for data on the polysphere $\mathbb{S}^{d_1} \times \cdots \times \mathbb{S}^{d_r}$. This test statistic will serve to

Figure 3: Dataset of $n = 177$ hippocampi ranked by the kernel density estimator of their shapes. The yellow–violet color gradient indicates the inward–outward ranking. Figure extracted from García-Portugués and Meilán-Vila (2024).

formally assess whether the components $\mathbf{X}_1, \ldots, \mathbf{X}_r$ of the polyspherical vector $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_r)$ supported on $\mathbb{S}^{d_1} \times \cdots \times \mathbb{S}^{d_r}$ are independent between them given an independent and identically distributed sample of $\mathbf{X}$. For example, the sample of $\mathbf{X}$ can be the $n = 177$ hippocampus from Figure 3 that live on $\mathbb{S}^2 \times \overset{168}{\cdots} \times \mathbb{S}^2$ or a subset thereof.

The work will extend a previous approach, only sketched, in García-Portugués et al. (2015) in terms of generality, level of detail, and computational expediency. The proposal will also extend the test of independence from García-Portugués et al. (2024) that, using a different approach based on the characteristic function, is designed only for data on $\mathbb{S}^1 \times \mathbb{S}^1$. The proposed test statistic will feature kernel density estimators on $\mathbb{S}^{d_1} \times \cdots \times \mathbb{S}^{d_r}$ recently considered in García-Portugués and Meilán-Vila (2024) to conform the following squared distance:

$$T_n = \sqrt{n h_1^{d_1} \cdots h_r^{d_r}} \int_{\mathbb{S}^{d_1} \times \cdots \times \mathbb{S}^{d_r}} \left(\hat{f}_{h_1,\ldots,h_r}(\mathbf{x}) - \hat{f}_{h_1}(\mathbf{x}_1) \cdots \hat{f}_{h_r}(\mathbf{x}_r) \right)^2 \mathrm{d}\mathbf{x}, \quad (1)$$

where $\hat{f}_{h_1,\ldots,h_r}$ is the polyspherical kernel density estimator for the sample of $(\mathbf{X}_1, \ldots, \mathbf{X}_r)$ and $\hat{f}_{h_j}$ is the marginal kernel density estimator for the sample of $\mathbf{X}_j$, $j = 1, \ldots, r$. The bandwidths $h_1, \ldots, h_r$ control the level of smoothness and are tunning parameters. Software for evaluating efficiently $\hat{f}_{h_1,\ldots,h_r}$ and $\hat{f}_{h_j}$ using numerically-stable procedures that scales well for high dimensions has been recently developed in García-Portugués and Meilán-Vila (2023) for computing $\hat{f}_{h_1,\ldots,h_r}$ in $(\mathbb{S}^2)^{168}$. This will facilitate the implementation of (1)

and other test statistics based on these kernel density estimators.

An alternative to the squared distance statistic (1) is a Jensen–Shannon divergence statistic. In the testing problem of comparing the equality of k groups, the Jensen–Shannon divergence statistic has been shown to have a crucial advantage of better scaling with dimensions $d_1, \ldots, d_r$ and r (García-Portugués and Meilán-Vila, 2024). Therefore, it is expected that it will also show a key advantage over the more traditional test statistic (1) within the independence testing framework.

The potential contributions of this project are as follows:

- Design of a L^2-based test statistic.
- Derivation of a closed-form expression for statistic (1) that does not involve multivariate integrals for specific types of kernels, such as von Mises–Fisher and others.
- Design of a Jensen–Shannon divergence test statistic adapted to testing independence.
- Design of a computationally efficient permutation-based resampling to calibrate both tests.
- Derivation of the asymptotic null distribution of the L^2 test statistic: $T_n \to \mathcal{N}(\mu_0, \sigma_0^2)$ in distribution as $n \to \infty$.
- Derivation of the power under Pitman alternatives of the L^2 test statistic: $T_n \to \mathcal{N}(\mu_0 + \Delta, \sigma_0^2)$ in distribution as $n \to \infty$.
- Illustration of the null and non-null asymptotic properties with a numerical experiment.
- Implementation of the testing procedures in R via Rcpp (Eddelbuettel and François, 2011) and RcppArmadillo (Eddelbuettel and Sanderson, 2014).
- Consideration of rule-of-thumb bandwidth selectors for both tests.
- Construction of a simulation study to evaluate the performance of the tests, explore their dependence on the bandwidths, and show their consistency against general alternatives.
- Application of the tests to select independent spokes in s-reps.
- Application of the tests to detect serial dependence in sequences of observations on the sphere, such as sunspots records or comet orbits measurements.
- Application of the tests to detect serial dependence in a sequence of observations on the higher-dimensional sphere, such as rotations.

3 Key references

The following are the key references to carry out the project:

- Rosenblatt and Wahlen (1992).
- García-Portugués et al. (2015).
- García-Portugués et al. (2024).
- García-Portugués and Meilán-Vila (2023).
- García-Portugués and Meilán-Vila (2024).

4 Possible venues for publication

Among others, and depending on the attained contributions in the project, these are some journals where the developed work could be published:

- *Annals of the Institute of Statistical Mathematics.*
- *Computational Statistics and Data Analysis.*
- *Journal of Nonparametric Statistics.*
- *Journal of Statistical Planning and Inference.*
- *Metrika.*
- *Statistical Papers.*

Other venues can be explored as long as the journal is widely recognizable by the statistical community.

5 Possible future work

The envisaged project could be the basis for further approaches, such as a k-sample test on the polysphere from the squared-loss perspective or extensions to the polycylinder setup $\mathbb{R}^q \times \mathbb{S}^{d_1} \times \overset{r}{\cdots} \times \mathbb{S}^{d_1}$ to tackle the fully s-reps case with potential applications to guided variable selection in classification or regression models.

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